

Lab subject 3

Diffraction and Interference

3.1 Theory (Read before exercise!)

Although we have often treated light as rays propagating in straight lines this far, this picture does not fully describe the range of optical phenomena that can be investigated within these experiments. There are a number of additional concepts that are needed to explain certain limitations of ray optics and to describe some of the techniques that allow us to analyze images and control the amplitude and direction of light. This section is a brief review of two important phenomena in physical optics, interference and diffraction.

3.1.1 Huygen's Principle

Light is an electromagnetic wave made up of many different wavelengths. Since light from any source (even a laser!) consists of fields of different wavelength, it would seem that it would be difficult to analyze their resultant effect. But the effects of light made up of many colors can be understood by determining what happens for a monochromatic wave (one of a single wavelength) then adding the fields of all the colors present. Thus by analysis of these effects for monochromatic light, we are able to calculate what would happen in non-monochromatic cases. Although it is possible to express an electromagnetic wave mathematically, we will describe light waves graphically and then use these graphic depictions to provide insight to several optical phenomena. In many cases it is all that is needed to get going.

An electromagnetic field can be pictured as a combination of electric (E) and magnetic (H) fields whose directions are perpendicular to the direction of propagation of the wave (k), as shown in Figure 3.1. Because the electric and magnetic fields are proportional to each other, only one of the fields need to be described to understand what is happening in a light wave. In most cases, a light wave is described in terms of the electric field. The diagram in Figure 3.1 represents the field at one point in space and time. It is the arrangement of the electric and magnetic fields in space that determines how the light field

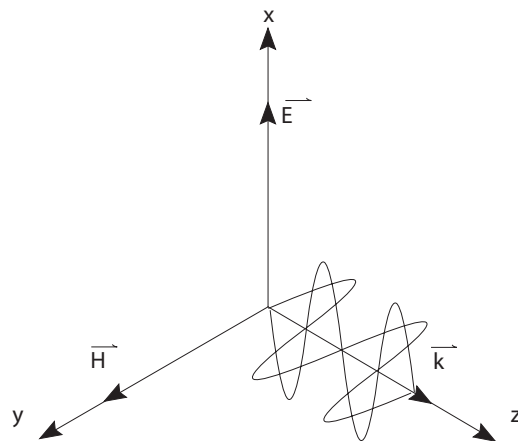


Figure 3.1: Monochromatic plane wave propagating along the z axis. For a plane wave, the electric field is constant in an $x-y$ plane. The vector $\vec{k}$ is in the direction of propagation.

progresses.

One way of thinking about light fields is to use the concept of wavefront. If we plot the electric fields as a function of time along the direction of propagation, there are places on the wave where the field is a maximum in one direction and other places where it is zero, and other places where the field is a maximum in the opposite direction, as shown in Figure 3.2. These represent different phases of the wave. Of course, the phase of the wave changes continuously along the direction of propagation. To follow the progress of a wave, however, we will concentrate on one particular point on the phase, usually at a point where the electric field amplitude is a maximum. If all the points in the neighborhood have this same amplitude, they form a surface of constant phase, or wavefront. In general, the wavefronts from a light source can have any shape, but some of the simpler wavefront shapes are of use in describing a number of optical phenomena.

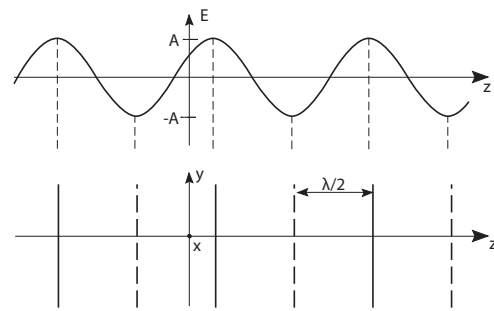


Figure 3.2: Monochromatic plane wave propagating along the z -axis. For a plane wave, the electric field is constant in an $x-y$ plane. The solid lines and dashed lines indicate maximum positive and negative field amplitudes.

A plane wave is a light field made up of plane surfaces of constant phase perpendicular to the direction of propagation. In the direction of propagation, the electric field varies sinusoidally such that it repeats every wavelength. To represent this wave, we have drawn the planes of maximum electric field strength, as shown in Figure 3.2, where the solid lines represent planes in which the electric field vector is pointing in the positive y -direction and the dashed lines represent plane in which the electric field vector is pointing in the negative y -direction. The solid planes are separated by one wavelength, as are the dashed planes.

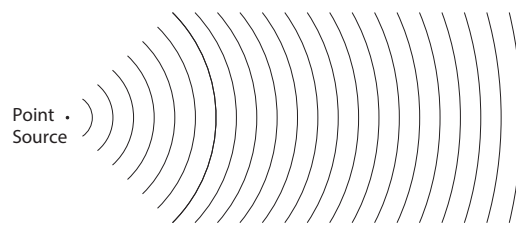


Figure 3.3: Spherical waves propagating outward from the point source. Far from the point source, the radius of the wavefront is large and the wavefronts approximate plane waves.

Another useful waveform for the analysis of light waves is the spherical wave. A point source, a fictitious source of infinitely small dimensions, emits a wavefront that travels outward in all directions producing wavefronts consisting of spherical shells centered about the point source. These spherical waves propagate outward from the point source with radii equal to the distance between the wavefront and the point source, as shown schematically in Figure 3.3. Far away from the point source, the radius of the wavefront is so large that the wavefronts approximate plane waves. Another way to create spherical waves is to focus a plane wave. Figure 3.4 shows the spherical waves collapsing to a point and then expanding. The waves never collapse to a true point because of diffraction (next Section). There are many other possible forms of wave fields, but these two are all that is needed for our discussion of interference.

What we have described are single wavefronts. What happens when two or more wavefronts are present in the same region? Electromagnetic theory shows that we can apply the principle of superposition: where waves overlap in the same region of space, the resultant field at that point in space and time is found by adding the electric fields of the individual waves at a point. For the present we are assuming that

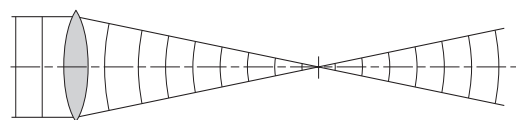


Figure 3.4: Generation of spherical waves by focusing plane waves to a point. Diffraction prevents the waves from focusing to a point.

the electric fields of all the waves have the same polarization (direction of the electric field) and they can be added as scalars. If the directions of the fields are not the same, then the fields must be added as vectors. Neither our eyes nor any light detector “sees” the electric field of a light wave. All detectors measure the square of the time averaged electric field over some area. This is the irradiance of the light given in terms of watts/square meter (W/m^2) or similar units of power per unit area.

Given some resultant wavefront in space, how do we predict its behavior as it propagates? This is done by invoking Huygen’s Principle. Or, in terms of the graphical descriptions we have just defined, Huygen’s Construction (see Figure 3.5): Given a wavefront of arbitrary shape, locate an array of point sources on the wavefront, so that the strength of each point source is proportional to the amplitude of the wave at that point. Allow the point sources to propagate for a time t , so that their radii are equal to ct (c is the speed of light) and add the resulting sources. The resultant envelope of the point sources is the wavefront at a time t after the initial wavefront. This principle can be used to analyze wave phenomena of considerable complexity.

3.1.2 Fresnel and Fraunhofer Diffraction

Diffraction of light arises from the effects of apertures and interface boundaries on the propagation of light. In its simplest form, edges of lenses, apertures, and other optical components cause the light passing through the optical system to be directed out of the paths indicated by ray optics. While certain diffraction effects prove useful, ultimately all optical performance is limited by diffraction, if there is sufficient signal, and by electrical or optical “noise”, if the signal is small.

When a plane wave illuminates a slit, the resulting wave pattern that passes the slit can be constructed using Huygens’ Principle by representing the wavefront in the slit as a collection of point sources all emitting in phase. The form of the irradiance pattern that is observed depends on the distance from the diffraction aperture, the size of the aperture and the wavelength of the illumination. If the diffracted light is examined close to the aperture, the pattern will resemble the aperture with a few surprising variations (such as finding a point of light in the shadow of circular mask!). This form of diffraction is called Fresnel (Freh-nell) diffraction and is somewhat difficult to calculate.

At a distance from the aperture the pattern changes into a Fraunhofer diffraction pattern. This type of diffraction is easy to calculate and determines in most cases, the optical limitations of most precision optical systems. The simplest diffraction pattern is that due to a long slit aperture. Because of the length of the slit relative to its width, the strongest effect is that due to the narrowest width. The resulting diffraction pattern of a slit on a distant screen contains maxima and minima, as shown in Figure 3.6(a). The light is diffracted strongly in the direction perpendicular to the slit edges. A measure of the amount of diffraction is the spacing between the strong central maximum and the first dark fringe in the diffraction pattern. The differences in Fraunhofer and Fresnel

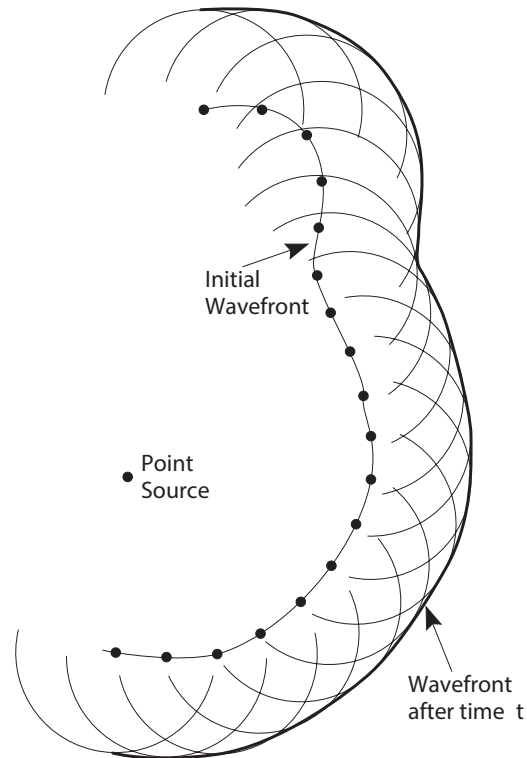


Figure 3.5: Huygen’s Construction of a propagating wavefront of arbitrary shape.

diffraction patterns will be explored in Exercise 3.3.

At distances far from the slit, the Fraunhofer diffraction pattern does not change in shape, but only in size. The fringe separation is expressed in terms of the sine of the angular separation between the central maximum and the center of the first dark fringe,

$$\sin \theta = \frac{\lambda}{w} \quad (3.1)$$

where w is the slit width and λ is the wavelength of the light illuminating the slit. Note that as the width of the slit becomes smaller, the diffraction angle becomes larger. If the slit width is not too small, the sine can be replaced by its argument,

$$\theta = \frac{\lambda}{w} \quad (3.2)$$

If the wavelength of the light illuminating the slit is known, the diffraction angle can be measured and the width of the diffracting slit determined. In Exercise 3.4 you will be able to do exactly this.

In the case of circular apertures, the diffraction pattern is also circular, as indicated in Figure 3.6(b), and the angular separation between the central maximum and the first dark ring is given by

$$\sin \theta = 1.22 \frac{\lambda}{D} \quad (3.3)$$

or for large D ,

$$\theta = 1.22 \frac{\lambda}{D} \quad (3.4)$$

where D is the diameter of the aperture. As in the case of the slit, for small values of λ/D , the sine can be replaced by its angle. The measurement of the diameter of different size pinholes is part of Exercise 3.3.

One good approximation of a point source is a bright star. A pair of stars close to one another can give a measure of the diffraction limits of a system. If the stars have the same brightness, the resolution of the system can be determined by the smallest angular separation between such sources that would still allow them to be resolved. This is provided that the aberrations of the optical system are sufficiently small and diffraction is the only limitation to resolving the images of these two point sources. Although it is somewhat artificial, a limit of resolution that has been used in many instances is that two point sources are just resolvable if the maximum of the diffraction pattern of one point source falls on the first dark ring of the pattern of the second point source, as illustrated in Figure 3.7, then

$$\theta_R = 1.22 \frac{\lambda}{D} \quad (3.5)$$

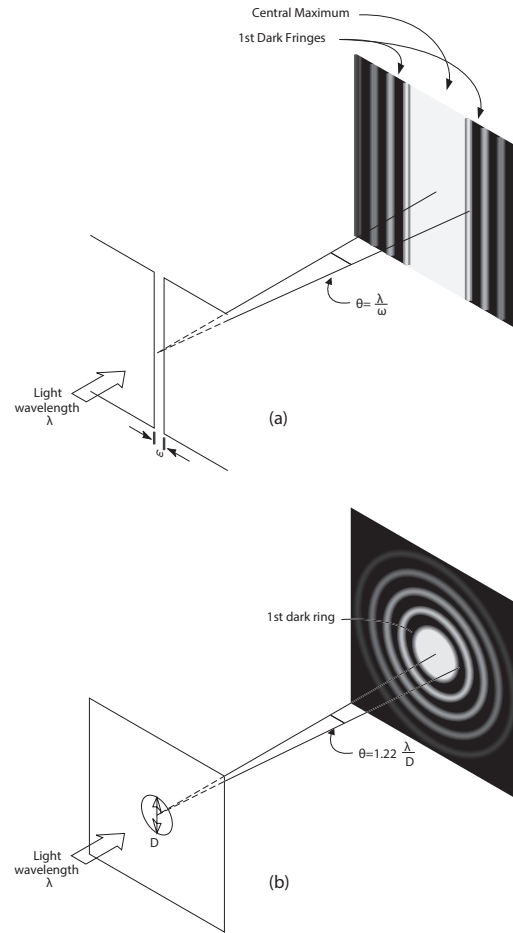


Figure 3.6: Diffraction of light by apertures. (a) Single slit. (b) Circular aperture.

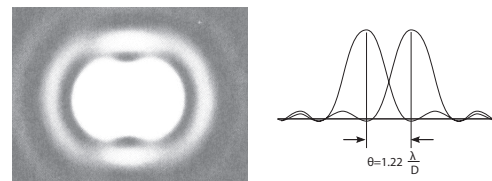


Figure 3.7: Rayleigh criterion. The plot of the intensity along a line between the centers of the two diffraction patterns is shown below a photo of two sources just resolved as specified by the Rayleigh criterion. (Photo by Vincent Mallette)

This condition for resolution is called the Rayleigh criterion. It is used in other fields of optical design, such as specifying the resolution of a optical systems.

3.1.3 Interference

While diffraction provides the limits that tells us how far an optical technique can be extended, interference is responsible for some of the most useful effects in the field of optics — from diffraction gratings to holography. As we shall see, an interference pattern is often connected with some simple geometry. Once the geometry is discovered, the interference is easily understood and analyzed.

3.1.4 Young's Experiment

In Figure 3.8 the geometry and wave pattern for one of the simplest interference experiments, Young's experiment, is shown. Two small pinholes, separated by a distance d , are illuminated by a plane wave, producing two point sources that create overlapping spherical waves. The figure shows a cross-sectional view of the wavefronts from both sources in a plane containing the pinholes. Notice that at points along a line equidistant from both pinholes, the waves from the two sources are always in phase. Thus, along the line marked C the electric fields always add in phase to give a field that is along the line, which is proportional to the square of the electric field, will be four times that due to a single pinhole. When electric fields add together to give a larger value it is referred to as constructive interference. There are other directions, such as those along the dotted lines marked D , in which the waves from the two sources are always 180° out of phase. That is, when one source has a maximum positive electric field, the other has the same negative value so the fields always cancel and no light is detected along these lines marked D , as long as both sources are present. This condition of canceling electric fields is called destructive interference. Between the two extremes of maximum constructive and destructive interference, the irradiance varies between four times the single pinhole irradiance and zero. It can be shown that the total energy falling on the surface of a screen placed in the interference pattern is neither more nor less than twice that of a single point source; it is just that interference causes the light distribution to be arranged differently!

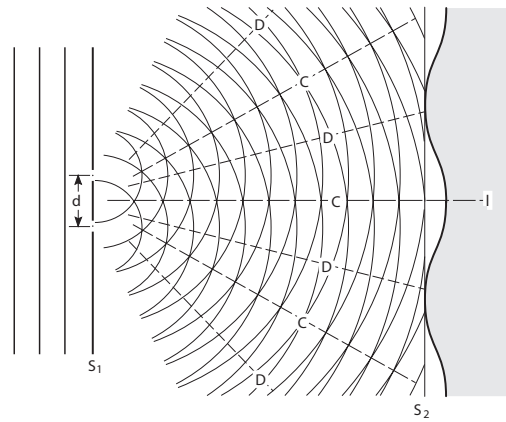


Figure 3.8: Young's Experiment. Light diffracted through two pinholes in screen S_1 spreads out toward screen S_2 . Interference of the two spherical waves produces a variation in irradiance (interference fringes) on S_2 that is plotted to the right of the screen.

Some simple calculations will show that the difference in distances traveled from pinholes to a point on the screen is

$$\Delta r = d \sin \theta \quad (3.6)$$

In the case of constructive interference, the wavefronts arrive at the screen in phase. This means that there is either one or two or some integral number of wavelength difference between the two paths traveled by the light to the point of a bright fringe. Thus, the angles at which the bright fringes occur are given by

$$\Delta r = d \sin \theta = n\lambda \quad (n = 1, 2, 3, \dots) \quad (3.7)$$

If the above equation is solved for the angles θ_n at which the bright fringes are found and one applies the approximation that for small angles the sine can be replaced by its angle in radians,

one obtains:

$$\theta_n = n\lambda/d \quad (n = 1, 2, 3, \dots) \quad (3.8)$$

The angular separation by neighboring fringes is then the difference between θ_{n+1} and θ_n :

$$\Delta\theta = \lambda/d \quad (3.9)$$

It is this angular separation between fringes that will be measured in Exercise 3.4 to determine the separation between two slits.

3.1.5 The Diffraction Grating

It is a somewhat confusing use of the term to call the item under discussion a diffraction grating. Although diffraction does indeed create the spreading of light from a regular array of closely spaced narrow slits, it is the combined interference of multiple beams that permits a diffraction grating to deflect and separate the light. In Figure 3.9 a series of narrow slits, each separated from its neighboring slits by distance d , are illuminated by a plane wave. Each slit is then a point (actually a line) source in phase with all other slits. At some angle θ_d to the grating normal, the path-length difference between neighboring slits will be (see inset to Figure 3.9)

$$\Delta x = d \sin(\theta_d) \quad (3.10)$$

Constructive interference will occur at that angle if the path-length difference Δx is equal to an integral number of wavelengths:

$$m\lambda = d \sin(\theta_d) \quad (m = \text{an integer}) \quad (3.11)$$

This equation, called the grating equation, holds for any wavelength. Since any grating has a constant slit separation d , light of different wavelengths will be diffracted at different angles. This is why a diffraction grating can be used in place of a prism to separate light into its colors. Because a number of integers can satisfy the grating equation, there are a number of angles into which monochromatic light will be diffracted. This will be examined in Exercise 3.4. Therefore, when a grating is illuminated with white light, the light will be dispersed into a number of spectra corresponding to the integers $m = \dots, \pm 1, \pm 2, \dots$, as illustrated in Figure 3.10(a). By inserting a lens after the grating, the spectra can be displayed on a screen one focal length from the lens, Figure 3.10(b). These are called the orders of the grating and are labeled by the value of m .

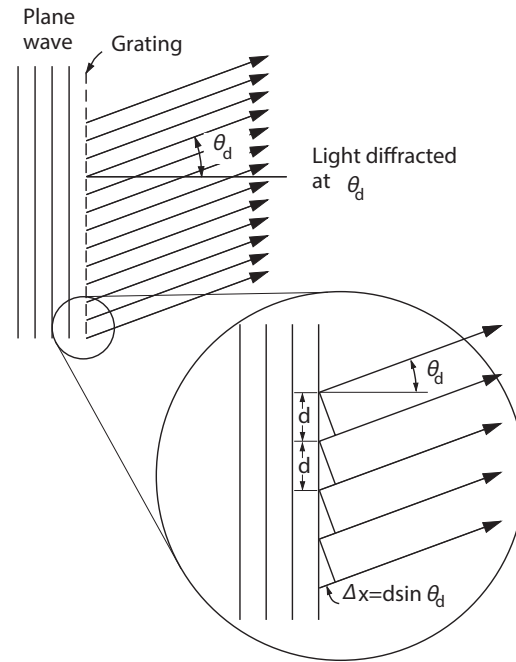


Figure 3.9: Diffraction of light by a diffraction grating.

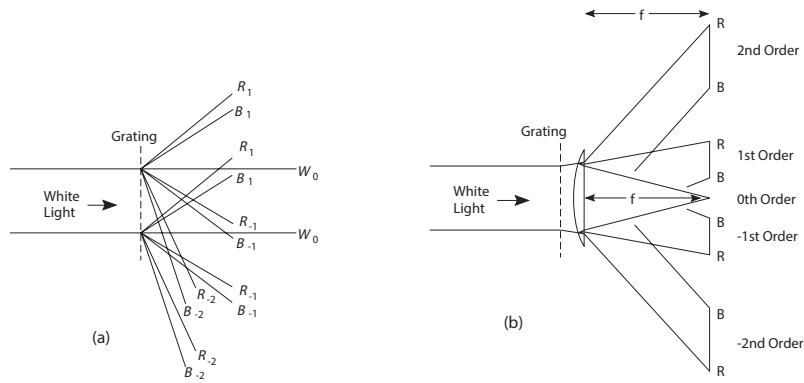


Figure 3.10: Orders of diffraction from a grating illuminated by white light. (a) Rays denoting the upper and lower bounds of diffracted beams for the red (R) and blue (B) ends of the spectrum; (b) spectra produced by focusing each collimated beam of wavelengths to a point in the focal plane.

3.2 Pre-lab numerical assignment: Diffraction

3.2.1 Single slit

Let D be the slit width, λ the wavelength, y the distance of the screen to the slit and x is a perpendicular axis to y along the screen. The expression of the intensity normalized to $I(0)$ for a single slit using the Fraunhofer approximation is:

$$\frac{I(x)}{I(0)} = \left[\frac{\sin\left(\frac{\pi D}{\lambda} \frac{x}{y}\right)}{\left(\frac{\pi D}{\lambda} \frac{x}{y}\right)} \right]^2 \quad (3.12)$$

Make a computer program (in your preferred computer language) that calculates the normalized intensity as a function of x considering a wavelength of $\lambda = 532 \text{ nm}$ (Nd:YAG green laser), a slit width of $D = 2 \text{ }\mu\text{m}$, and a distance between the slit and the screen of $x = 1 \text{ m}$. Plot your results.

Now, fix the value of λ while gradually increasing the value of D . What do you observe? Why do you think this happens? Finally, fix the value D of while gradually increasing the value of λ . Explain your observations.

After making the lab exercise, how did your numerical results compare to your observations in the lab? Was there an important difference?

Hand in a print of your computer program and plots of your examples as pdf, together with your observations from each case. You can work together in teams, but each student delivers individually.

3.2.2 Double slit (Young's experiment)

Let D be the slit width, l the separation between slits, λ the wavelength, y the distance of the screen to the slit and x is a perpendicular axis to y along the screen. The expression of the intensity normalized to $I(0)$ for the Young's experiment using the Fraunhofer approximation is:

$$\frac{I(x)}{I(0)} = \left[\frac{\sin\left(\frac{\pi D}{\lambda} \frac{x}{y}\right)}{\left(\frac{\pi D}{\lambda} \frac{x}{y}\right)} \right]^2 \left[\cos\left(\frac{\pi l}{\lambda} \frac{x}{y}\right) \right]^2 \quad (3.13)$$

Make a computer program (in your preferred computer language) that calculates the normalized intensity as a function of x considering a wavelength of $\lambda = 532 \text{ nm}$ (Nd:YAG green laser), a slit width of $D = 2 \text{ }\mu\text{m}$, a separation between slits of $l = 4 \text{ }\mu\text{m}$, a number of slits of $N = 3$, and a distance between the slit and the screen of $x = 1 \text{ m}$. Plot your results.

Now, fix the values of λ and D while gradually increasing the value of l . Explain your observations.

After making the lab exercise, how did your numerical results compare to your observations in the lab? Was there an important difference?

Hand in a print of your computer program and plots of your examples as pdf, together with your observations from each case. You can work together in teams, but each student delivers individually.

3.2.3 Diffraction grating

Let D be the slit width, l the separation between slits, N the number of slits, λ the wavelength, y the distance of the screen to the slit and x is a perpendicular axis to y along the screen. The expression of the intensity normalized to $I(0)$ for a diffraction grating using the Fraunhofer approximation is:

$$\frac{I(x)}{I(0)} = \left[\frac{\sin(\frac{\pi D}{\lambda} \frac{x}{y})}{(\frac{\pi D}{\lambda} \frac{x}{y})} \right]^2 \left[\frac{\sin(N \frac{\pi l}{\lambda} \frac{x}{y})}{N \sin(\frac{\pi l}{\lambda} \frac{x}{y})} \right]^2 \quad (3.14)$$

Make a computer program (in your preferred computer language) that calculates the normalized intensity as a function of x considering a wavelength of $\lambda = 532 \text{ nm}$ (Nd:YAG green laser), a slit width of $D = 2 \text{ }\mu\text{m}$, a separation between slits of $l = 4 \text{ }\mu\text{m}$, a number of slits of $N = 3$, and a distance between the slit and the screen of $x = 1 \text{ m}$. Plot your results.

Now, fix the values of λ , D and l while gradually increasing the value of N . What do you observe? Why do you think this happens?

After making the lab exercise, how did your numerical results compare to your observations in the lab? Was there an important difference?

Hand in a print of your computer program and plots of your examples as pdf, together with your observations from each case. You can work together in teams, but each student delivers individually.